

2.2 Data Type: Primary data

- Info collected for the first time specifically for a specific research question
- Company-owned vs. Commission research company
- Pro:
 - Provide richer, more detailed info, answer a specific question
 - Current data
- Con:
 - Cost more
 - Take longer to collect

Demographics/socioeconomic:

- age, sex, income, marital status, occupation

Awareness/knowledge:

- Facts about products, price, usage

Motivation:

- why people buy (needs, wants, ideal self)

Attitudes/opinions:

- preference, feelings, opinions

Lifestyle:

- activities, interests, personality traits

Intentions/behaviors:

- intention to purchase, frequency, use behavior, timing

Two forms of data

1. Quantitative data: data that can either be counted or compared on a numeric scale
2. Qualitative data: data describes characteristics, qualities, and appears in descriptive words that can be examined for patterns or meanings.

Qualitative/Subjective Data

Feelings, characteristics, sensory details.

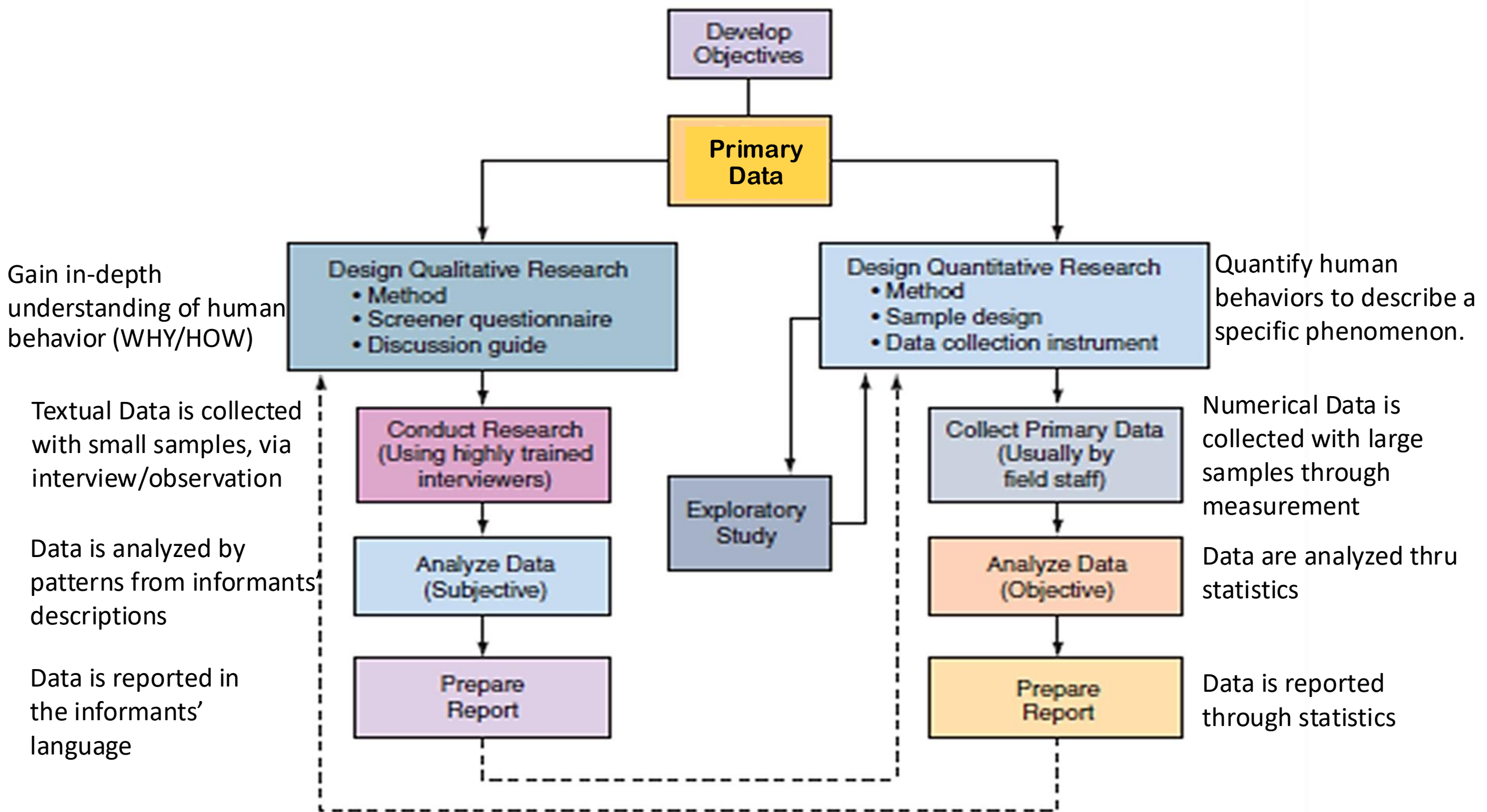
Answers "Why?" and "How?"

Quantitative/Objective Data

Price, rating, frequency, time, and other measurable metrics.

Answers "How much?" and "How many?"





Qualitative vs. Quantitative Research

Explore new/under-examined phenomenon

- Under what circumstances do shoppers feel deferent towards the salesperson

Explain the phenomenon (seek to understand)

- Why people buy counterfeit, second-hand goods?
- Why do you feel Kreng-jai?

Describe the experiences

- Consumers perception/feeling/experience towards the brand, relationship with the brand
- Consumer process (purchase, evaluation, product modification, cope up with new technology)

Descriptive research:

- How often, how much, how many, what percent?

Comparative research:

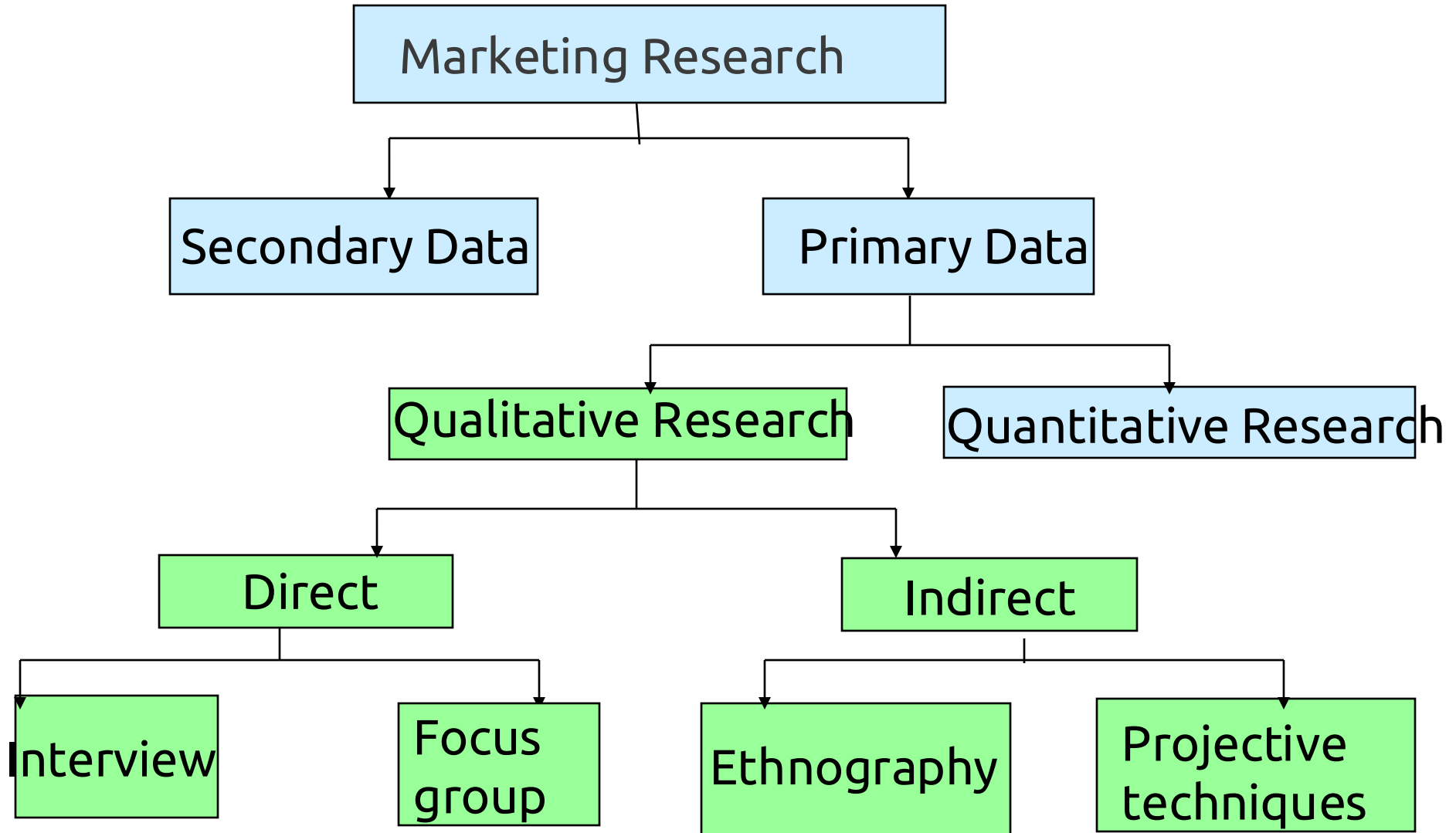
- What are the differences in perceptions towards Internet banking security between adolescents and pensioners?

Causal research (Relationship among variables)

- What is the relationship between body of others and food choice?
- Factors influencing airline passengers' satisfaction
- The impact of gender on consumer attitudes towards music piracy

Qualitative Research Methods

Research methodology that provide an in-depth understanding of human behavior and the reasons that govern such behavior. It investigates HOW and WHY questions



1) Focus Group



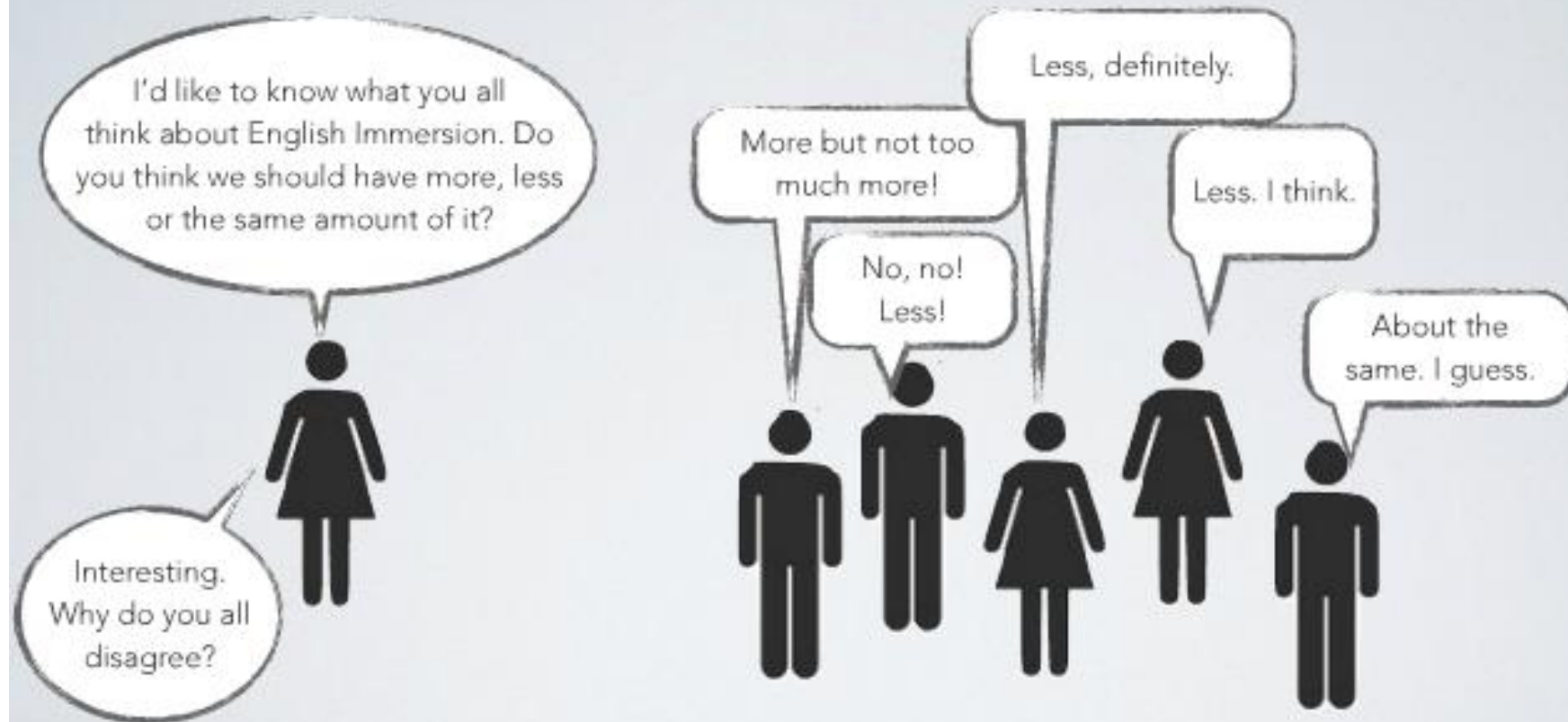
Group size	6-10 people
Group composition	Homogeneous, pre-screened Group dynamics
Physical setting	Relaxed, informal atmosphere
Time duration	1-3 hours
Recording	Use of audiocassettes and videotapes
Moderator	Observational, interpersonal and communication skills

FOCUS GROUPS: ADVANTAGES

Stimulation

Synergy

Save
time/cost



The whole is greater than the sum of its parts

FOCUS GROUPS: DISADVANTAGES

Difficult to get people in the same room

Difficult to control conversation

Huge amount of data



Dominant personalities

Follow others, not give true opinion

Misuse

2) In-Depth Interview

- A detailed, one-on-one conversation session
- Method for in-depth probing of personal opinions, beliefs, motivations, and values
- Last between 30-60 mins
- Unstructured (or loosely structured)
- Questions include
 - Behaviors: what a person has done
 - Knowledge: facts about the topic
 - Opinions: what a person thinks
 - Feelings: what a person feels
 - Sensory: what a person has seen, touched, heard, tasted, smelled
 - Background/demographics

Interview Guide

(page 1 of 2)

Icebreaker Questions • Tell me about yourself. • How would others describe you? • What are your strengths/weaknesses?	Notes:
General Interview Questions • Why did you leave your last job? • Tell me about any other jobs you've held. What were some of the duties/responsibilities? • What did you like the most/least about your last job? • What past work experience has best prepared you for this job?	
Customer Service • Tell me about a time you handled an angry customer. • Tell me about a time you went above and beyond to help a customer. • What does customer service mean to you? Why is it important?	
Attention to Detail • Tell me about a time when you caught an error that others had missed. • Tell me about a time when you did not pay as close attention to something as you should have.	

Interview

Advantage

- Rich depth of info
- Uncover hidden issues
- Suit for personal/sensitive issue
- Flexible
 - Allow the interviewee to elaborate.
 - Allow the interviewer to ask follow-up questions
- No social pressure on respondents to conform

Disadvantage

- Time consuming
- Expensive
- Require skilled interviewers
- Interviewer bias

Interviewing techniques

1. Probing technique: Ask follow-up questions

Request for elaboration: "Tell me more about that." "Give me an example of"

Request for definition: "What do you mean by...?" "What does the term _____ mean to you?"

Request for word associations: "What other word(s) do you link with _____?" "Give me some synonyms that also describe _____."

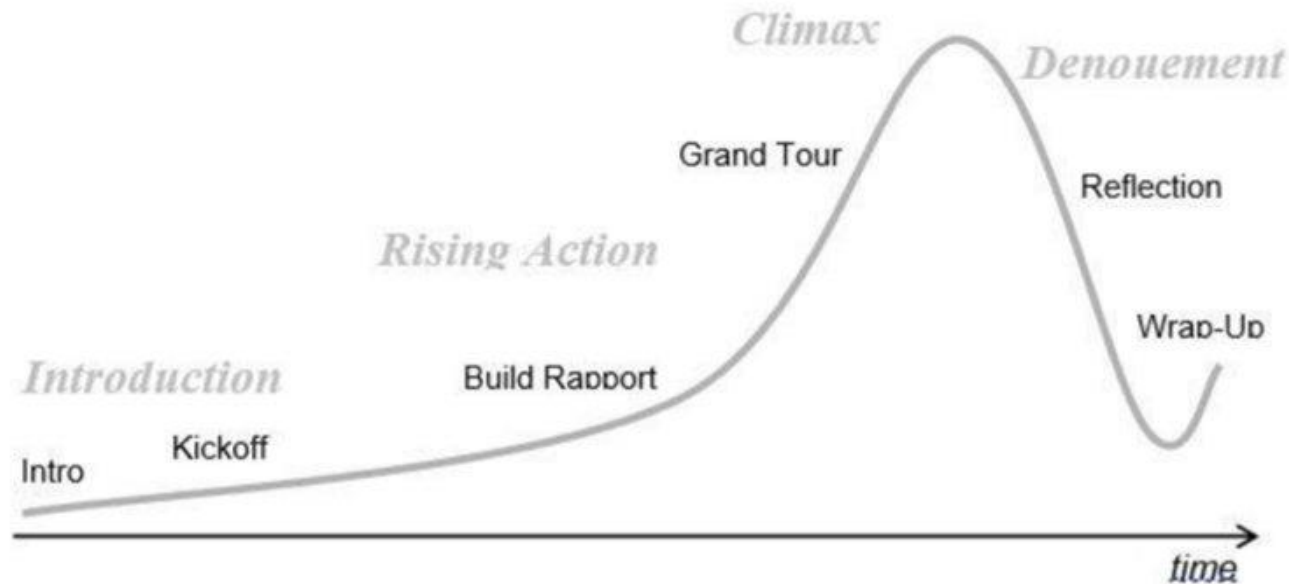
Request for clarification: "How does that differ from"
"In what circumstances do you.....?"

Request for comparison: "How is _____ similar to _____?"
"Which costs more, X or Y?"

Request for classification: "Where does _____ fit?" "What else is in the category of _____?"

"Silent" probe: This is a non-verbal probe and is characterized by such actions as raised eyebrows or hand gestures such as moving the right hand in a rolling motion that signifies "Tell me more."

Interview often flows like a good story:



- I: Why do you buy **BMW**?
- R: Because it has good performance
- I: What is it that you like about good performance?
- R: It gives me the high speed drive
- I: What is it about high speed drive?
- R: It gives me a sense of control
- I: Why do you want to be in control?
- R: I feel powerful
- I: Why do you want to feel powerful?
- R: I want to be a **leader**

2. Laddering technique

To discover end benefits, researcher asks about a specific product feature and poses a series of why questions that builds on previous answers.

- Which feature do you like best?
- What does that feature do? (functional benefit)
- What does the functional benefit do for you? (higher benefit)
- What does the benefit do for you? (emotional benefit)



Implication: BMW is related to leadership

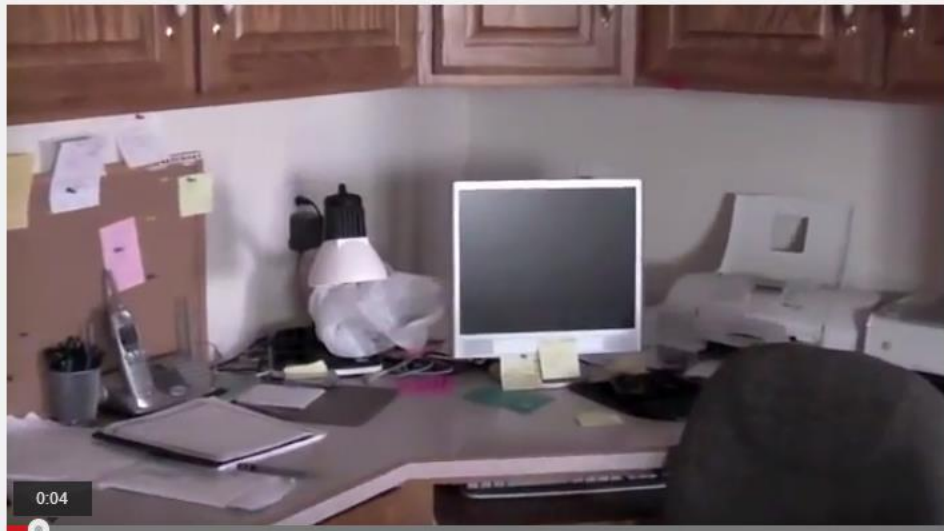
3) Ethnography

What consumers say they do ≠ what they actually do.

- The branch of anthropology that tries to understand how people live their lives by observing & interacting with consumers while they live with, shop for, and use a product or service
- Observe in their natural settings.
 - Visit their homes
 - How they live their everyday lives. Their activities.
 - What they value in their lives.
 - How they use products
 - Watch people shopping in the retail setting
 - How they walk and move. How they choose brands, features
 - Problems they encounter.

What we observe?

Phenomenon	Example
Physical action	Shoppers movement pattern in a store
Verbal behavior	Conversations
Expressive behavior	Facial expression, tone voice, body languages
Spatial	How close customers stand to product
Temporal patterns	How long customers wait for their order
Physical object	What items are stored in consumers' pantries
Internet activity	Websites viewed, time spent, social networking habits



Shop Along (Interview + Observe)

- A researcher shops alongside the actual consumer, asking questions to respondents as he/she shops
- Think-aloud protocol
- Explore how and why consumers shop the way they do
 - Shopping process, Influential factors
- Pro:
 - Capture real-time experiences.
 - Help uncover shopping influence
- Con:
 - Not shop as they really do



Example



- P&G researchers observe shoppers in a store , or in a consumer's home.
- When observing consumers using its Tide laundry detergent in India, P&G noticed that because clothes were often washed by hand, the detergent sometimes caused skin irritation.
- P&G introduced Tide Naturals, which cleaned well without causing irritation.

- Use ethnography to uncover the likes/dislikes of boys (6-14)
- Boys, less open than girls in regular surveys
- Ethnographers visit their homes
 - Observe their rooms, hobbies
 - Accompany them and mom to see where they go, what they are interested in, and what they buy
 - Spend time with them for 18 months
- Results
 - Created a new TV channel, Disney XD, tailored toward the interests of boys.
 - A 10% increase in male viewership

Ethnography: Pro & Con

Advantage

- Data are recorded when actual and nonverbal behavior takes place
- Data are free from distortions, inaccuracies, and other response biases (e.g. self-report bias)
- Uncover problems customers may not know

Disadvantage

- Time consuming
- Very costly

Netnography = Ethnography on the Internet

- Immerse in consumers' world and a consumer network and gain unbiased, unobtrusive consumer insights
- Source: Discussion board, rating and review sites, e-commerce websites, consumer weblogs, social media group
 - Relevant, traffic, rich data, member interaction
- Data: culture, language, rituals, meanings, practices, behaviors
- Technique: Start acting like a new member (not researcher) and contributing
- Pro:
 - Gain deeper, more accurate insider knowledge
 - Obtain data quickly, cheaply and in a timely manner
- Con
 - Identity validation, reliability and integrity of information
 - Loss of non-verbal (tone of voice, facial expression)

Netnography Case Study: Modern Foldable Smartphones (e.g., Z Flip)



1. COMPETITOR / SWOT ANALYSIS

Summary: Comparison to standard flagships; cameras & battery lacking for price.

“For \$1,000, I can't accept these cameras. My 3-year-old iPhone takes better low-light photos.”

“It's the coolest phone... but the battery anxiety is real compared to my old Ultra.”



3. TREND FORECASTING (Nostalgia & Fashion)



Summary: Phone as fashion accessory & 2000s nostalgia trip.

“Hanging up... by snapping the phone shut is the most satisfying feeling on earth. 2004 vibes.”

“I bought it because it fits in my tiny purse and looks cute... specs don't matter.”



2. PRODUCT DESIGN (The 'Crease')

Summary: Dominant conversation about visible screen crease barrier.

“Does the crease annoy you when scrolling TikTok? It's all I can see...”

“You stop noticing it after 2 days, I promise. It's like the notch...”



4. USABILITY (Durability Anxiety)

Summary: Extreme fear of breaking fragile inner screen.

“I have 'Fold Anxiety.' Every time I open it, I'm terrified I'll hear a crack.”

“Don't take it to the beach. One grain of sand in that hinge and it's game over.”



Netnography Analysis:
Mining Foldable Phone Communities

Netnography Case Study: The Listerine Experience



1. COMPETITOR / SWOT ANALYSIS

Summary: Not just mouthwash; competes with specialized treatments (e.g., antifungal, anti-itch).

"Cheaper than athlete's foot cream, and it actually works."

"Why buy bug spray when I have this under the sink?"



2. PRODUCT USAGE (OFF-LABEL)

Summary: Discovering unconventional, user-generated applications.

"Soaking my feet in it for 20 mins... fungus gone!"

"Dabbed it on a mosquito bite, itch stopped instantly."



3. TREND FORECASTING (DIY HEALTH)



Summary: Rising trend of home remedies and 'life hacks'.

"Old-school remedy, but it works better than new stuff."

"My grandma swears by it for dandruff, trying it tonight."



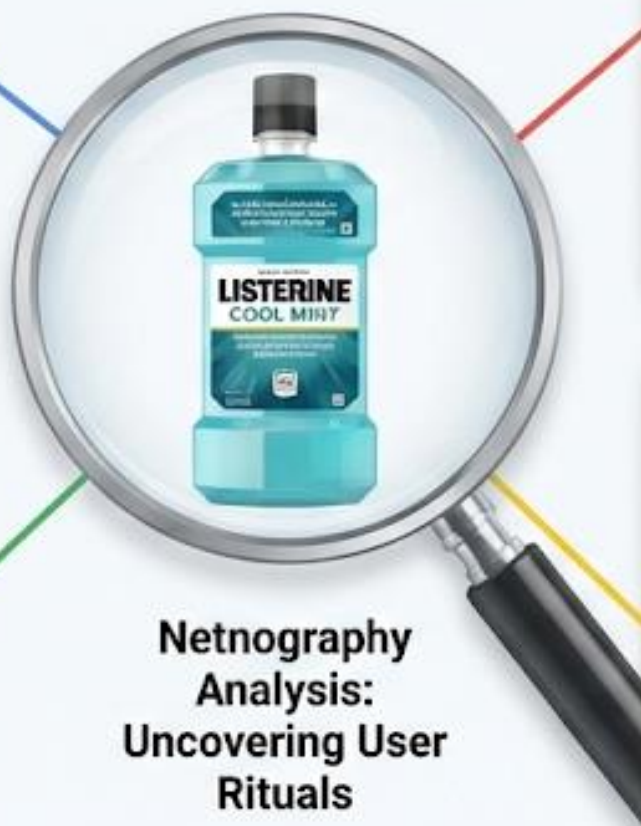
4. USABILITY & SENSATION



Summary: Intense sensation is polarizing but seen as effective.

"Burns like crazy, but that means it's killing the germs!"

"Can't handle the taste for 30 seconds, but my breath is nuclear fresh."

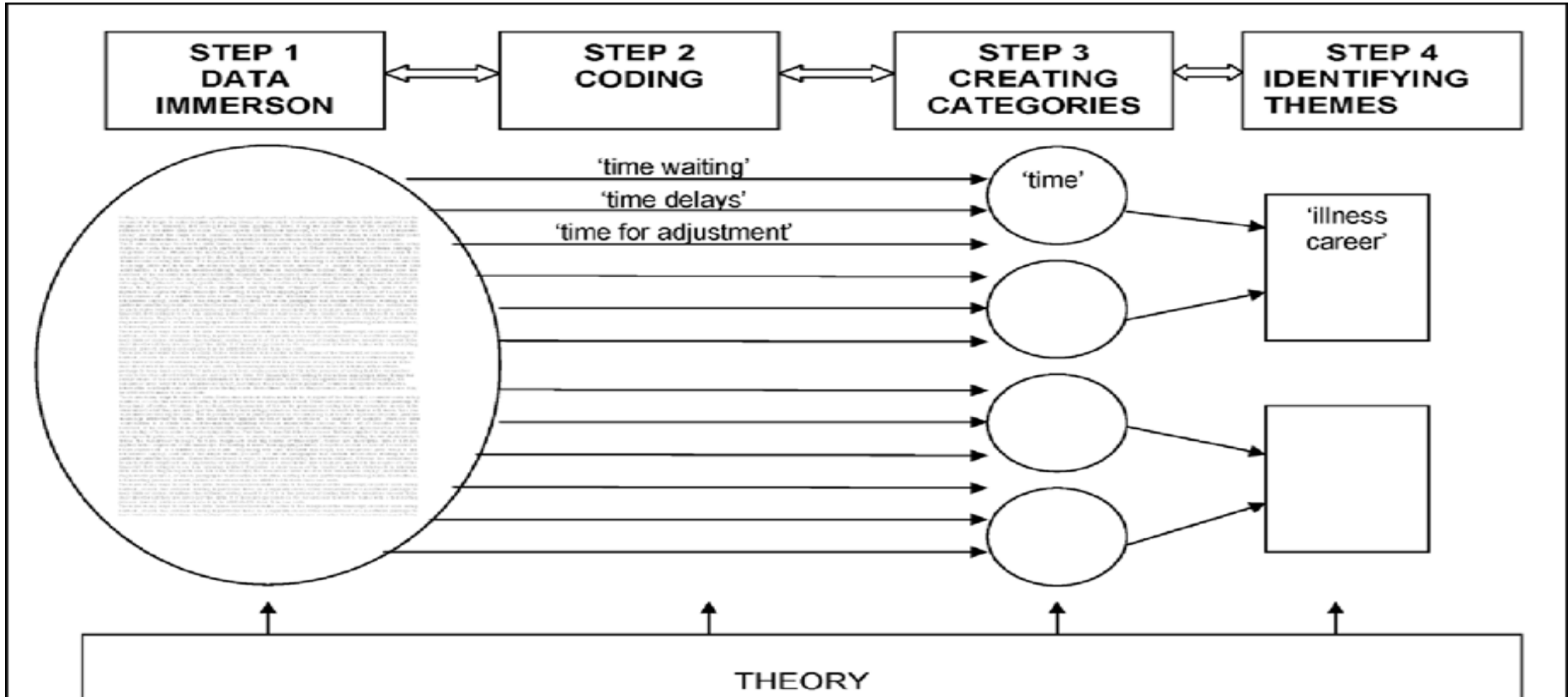


**Netnography Analysis:
Uncovering User Rituals**

Step 3: Qualitative Data Collection

- Recruit the sample. Contact in advance.
- Tell them instructions, duration, compensation
- Recruit field workers. Brief and control them.
- **Fact (consumer's perspectives)**
 - Record text/audio/video/photo (ask permission)
 - Respondents' diaries (in advance)
- **Opinion (researcher's interpretation)**
 - Field note
 - Description: Time, place, people, surroundings
 - Analysis: What do you learn?
 - Reflection: What do you feel?

Step 4-5: Qualitative Data Analysis & Presentation





Insights Database

Board ▾

Participant 1 1

Insight title

[Paste quote here]

Participant 1

High importance

Epic 1

Participant 2 1

Insight title

[Paste quote here]

Participant 2

Theme 11

High importance

Participant 3 1

Insight title

[Paste quote here]

Participant 3

Theme 2

Medium importance

Participant 4 1

Insight title

[Paste quote here]

Participant 4

Theme 9

Low importance

- Transcribe data, Read & re-read the data, hi-light and code the quotes

- Extract the highlight into a database for grouping codes and creating relations between codes

Date	time	Channel	Type	Topic	Link	Bank	Category	Sub Category	Sentiment	Topics
8/1/2017	1:15:11 PM	TWITTER	post	"Kbank app... Kbank's Net Banking is probably the easiest to use lol. Love it, money moves fast."	http://twitter.com/_HE4R4th/status/892267446735781888	KBANK	Service channel	Mobile Banking	Positive	System/Process/Policy
8/1/2017	3:34:58 PM	TWITTER	post	Totally broke. Took all my money to buy GSB Savings Lottery."	http://twitter.com/ameynniar/status/892302623956639744	GSB	Saving	Saving account	Positive	Product
8/1/2017	8:39:55 AM	TWITTER	post	I really want TMB All Free. Damn, staying here I must have paid thousands in withdrawal fees."	http://twitter.com/AniwatVitetas/status/892198175590711297	TMB	Transaction	Cards	Positive	Fee
8/1/2017	12:41:22 PM	TWITTER	post	"Queue* at SCB today. Walked to pay BBL credit card and buy lip balm, still made it back in time to wait."	http://twitter.com/CatsRSoft/status/892258937726160896	SCB	Service channel	Branch	Negative	Service

Step 5: Presentation

TABLE 1
Group and Category Classification by Type of Incident Outcome

Group and Category	Type of Incident Outcome				Row Total	
	Satisfactory		Dissatisfactory			
	No.	%	No.	%	No.	%
Group 1. Employee Response to Service Delivery System Failures						
A: Response to unavailable service	24	6.9	29	8.2	53	7.6
B: Response to unreasonably slow service	17	4.9	53	15.1	70	10.0
C: Response to other core service failures	40	11.5	69	19.6	109	15.6
Subtotal, group 1	81	23.3	151	42.9	232	33.2
Group 2. Employee Response to Customer Needs and Requests						
A: Response to "special needs" customers	36	10.4	6	1.7	42	6.0
B: Response to customer preferences	51	14.7	37	10.5	88	12.6
C: Response to admitted customer error	20	5.8	8	2.3	28	4.0
D: Response to potentially disruptive others	7	2.0	4	1.1	11	1.6
Subtotal, group 2	114	32.9	55	15.6	169	24.2
Group 3. Unprompted and Unsolicited Employee Actions						
A: Attention paid to customer	48	13.				
B: Truly out-of-the-ordinary employee behavior	22	6.				
C: Employee behaviors in the context of cultural norms	16	4.				
D: Gestalt evaluation	55	15.				
E: Performance under adverse circumstances	11	3.				
Subtotal, group 3	152	43				
Column Total	347	49				

Group 1 Sample Incidents: Employee Response to Service Delivery Failures

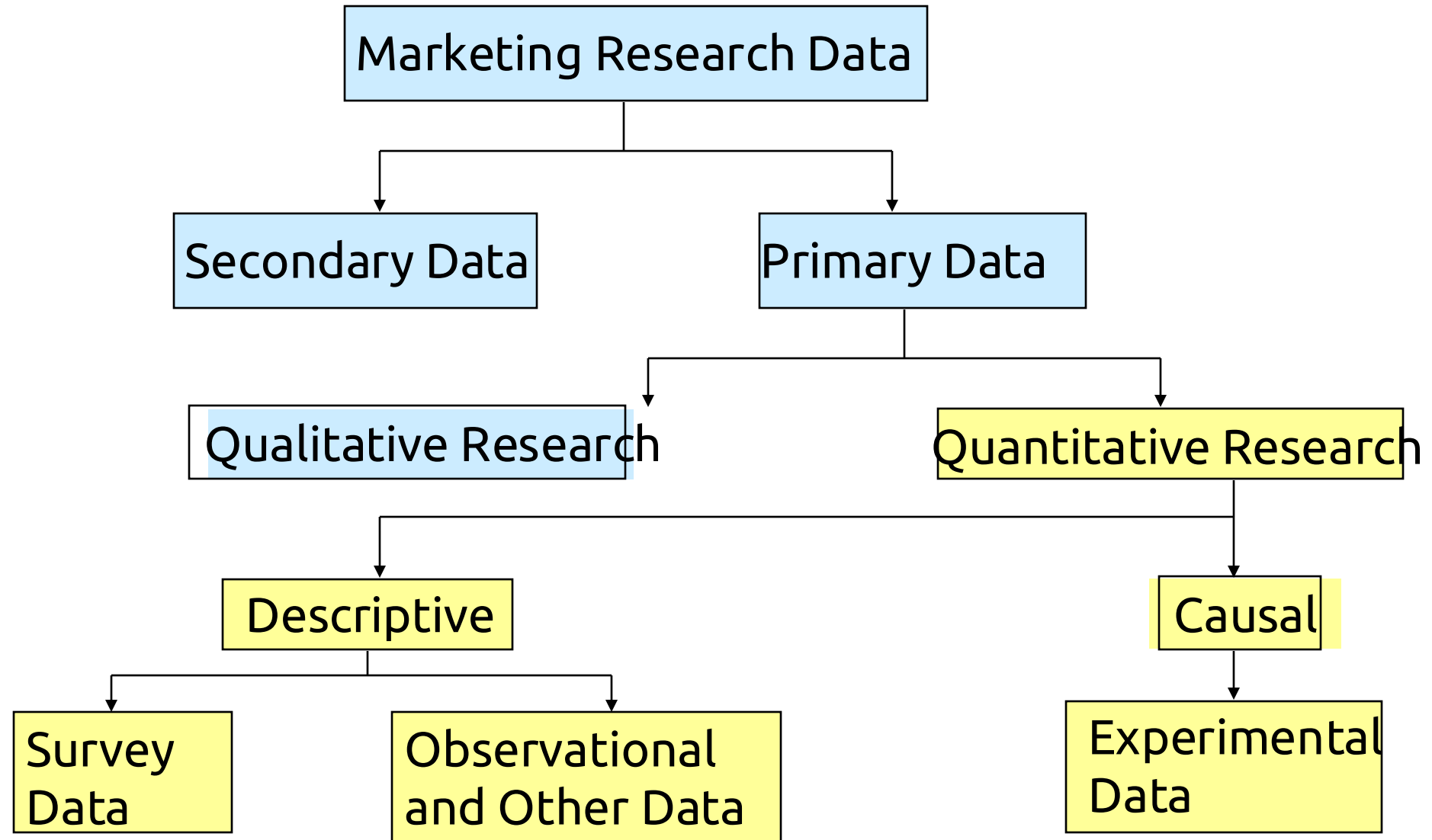
	Incident	
	Satisfactory	Dissatisfactory
A. Response to Unavailable Service	They lost my room reservation but the manager gave me the V.P. suite for the same price.	We had made advance reservations at the hotel. When we arrived we found we had no room—no explanation, no apologies, and no assistance in finding another hotel.
B. Response to Unreasonably Slow Service	Even though I didn't make any complaint about the hour and a half wait, the waitress kept apologizing and said that the bill was on the house.	The airline employees continually gave us erroneous information; a one-hour delay turned into a six-hour wait.
C. Response to Other Core Service Failures	My shrimp cocktail was half frozen. The waitress apologized, and didn't charge me for any of my dinner.	One of my suitcases was all dented and looked as though it had been dropped from 30,000 feet. When I tried to make a claim for my damaged luggage, the employee insinuated that I was lying and trying to rip them off.

Organizing qualitative data
into a frequency table

Presenting categories with
quotes that represent each
category

Quantitative Research Methods

Research methodology that seeks to quantify the data via some form of statistical analysis to investigate observable phenomena



Quantitative Observation

91% of adults say they always wash their hands after using a restroom.
But observation reveals only 77% actually do

- A systematic record of the behavioral patterns of people, objects, events.
- Involve the measurement of quantity or amount.
- In most cases, observer does not question or communicate with the people being observed
- Different types of observation
 - Disguised vs. non-disguised: the person not know vs. know they are being observed
 - Personal observation:
 - Field workers record traffic counts/flows in a store
 - Mechanical observation:
 - TV meter, video camera, scanner, eye-tracking, voice pitch analyzer
 - Indirect observation (observe the result, not the action)
 - words, sentiments, conversations, length of discussion about brand from forum posts, tweets

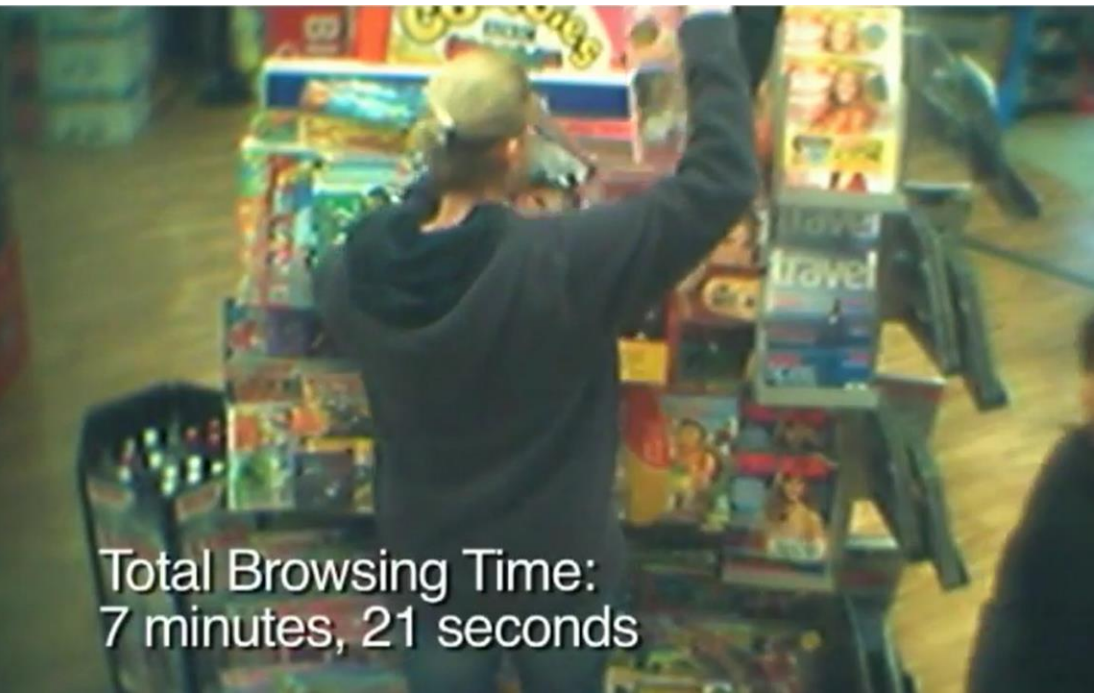
In-store Observation

Trackers record

- age, sex and appearance, number of customers
- items they touch
- amount of time they look at, touch, read product description, before purchase
- amount of time they spent at each station



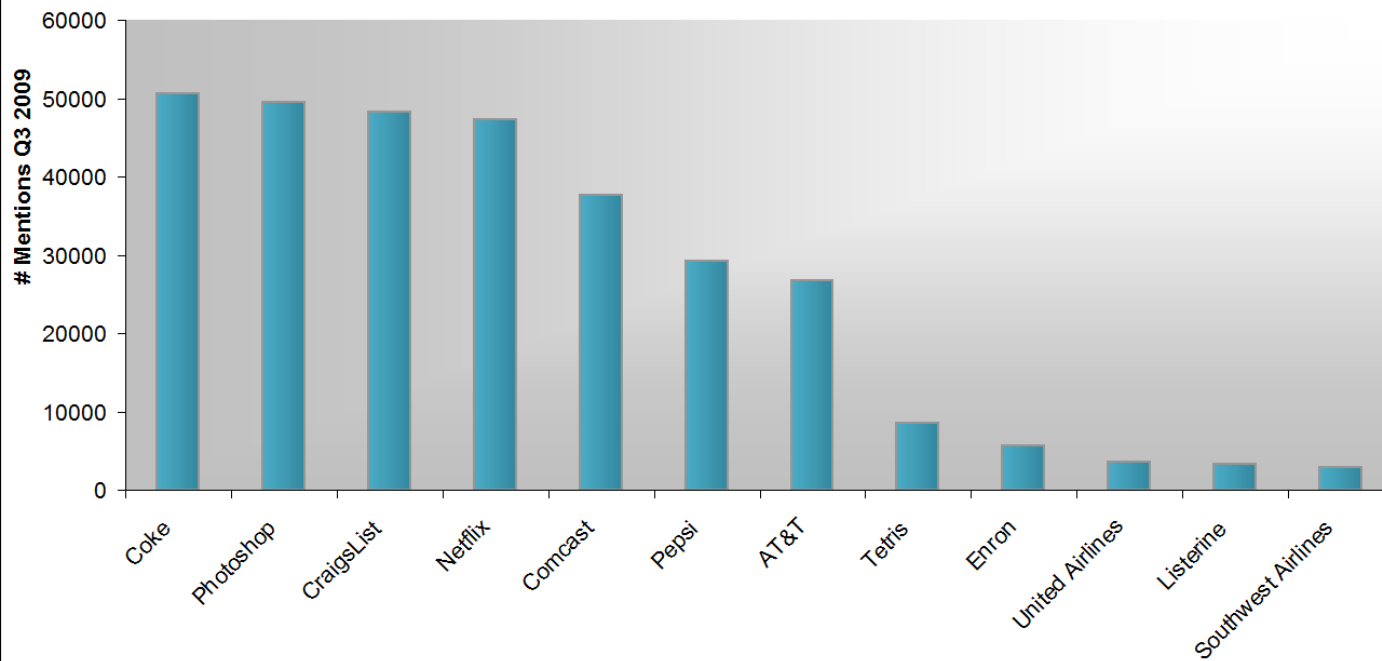
Rimmel lipsticks were the most popular product. Out of the 16 people looking at lipsticks, 11 looked at Rimmel



Total Browsing Time:
7 minutes, 21 seconds



Mentions per Brand



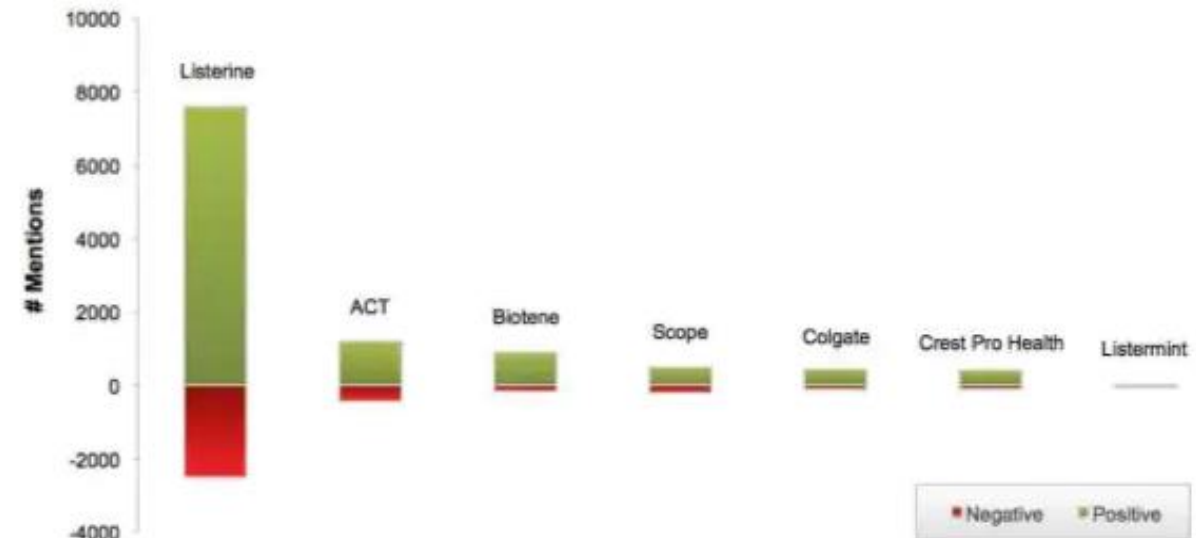
Social listening

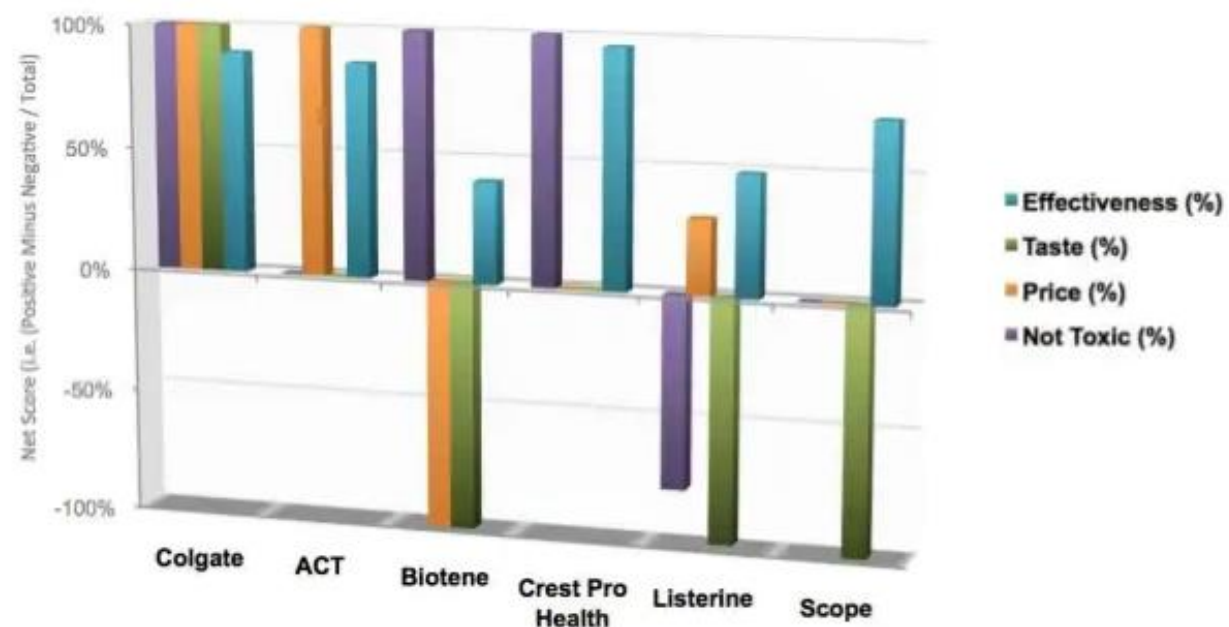
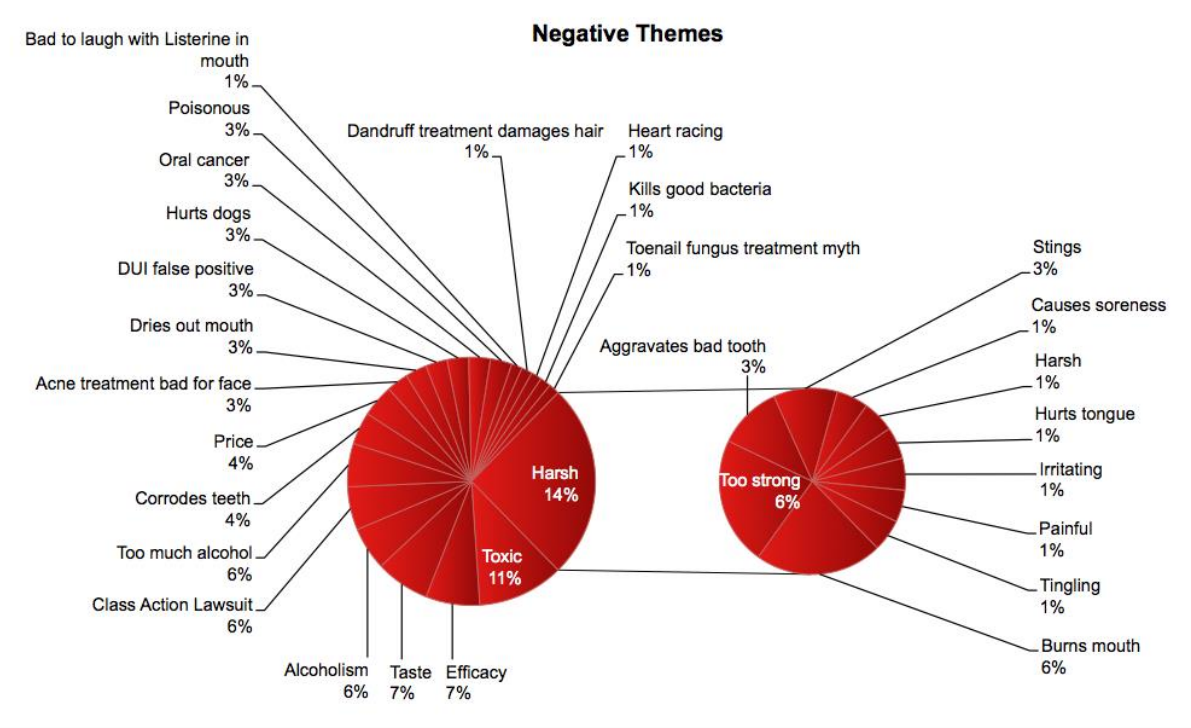


Quantitative analysis

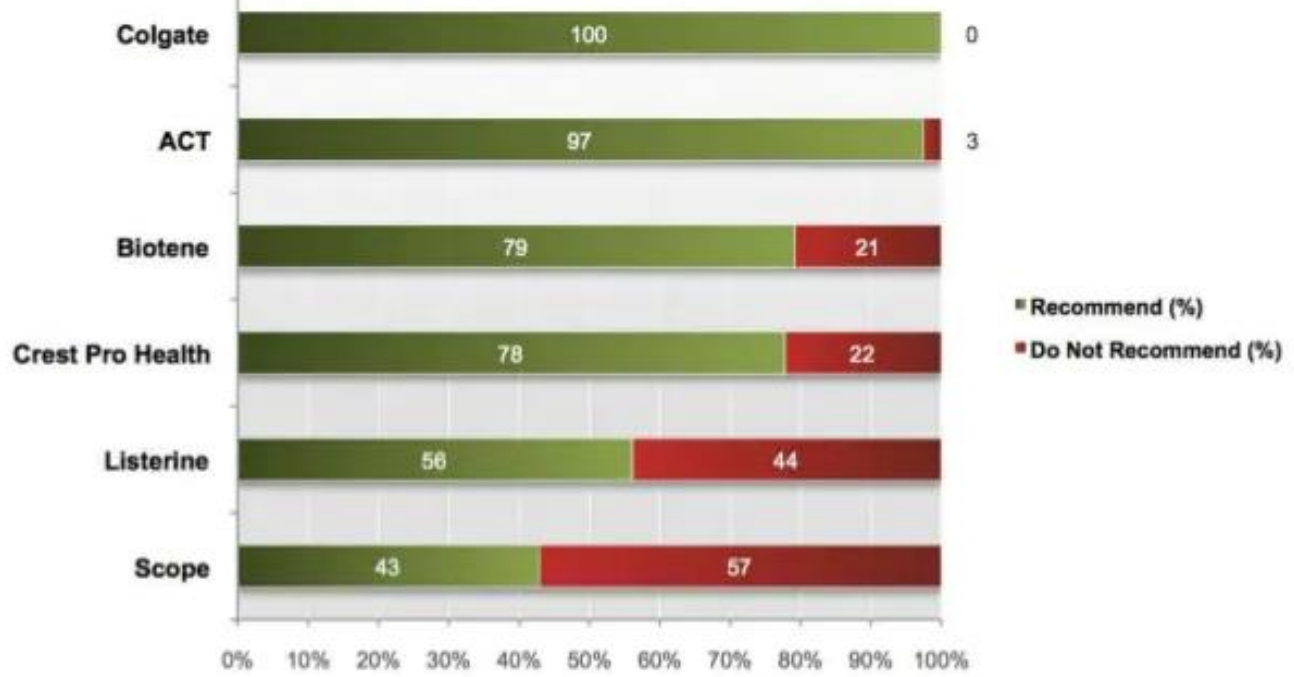
- amount of discussion about Listerine relative to other topics
- consumer sentiment and passion about Listerine relative to other topics

Sentiment by Brand





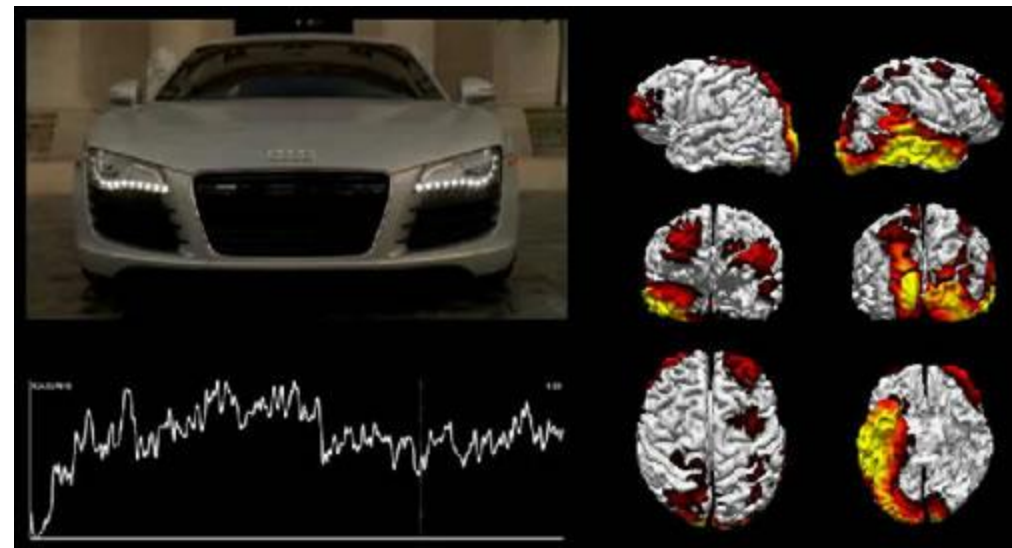
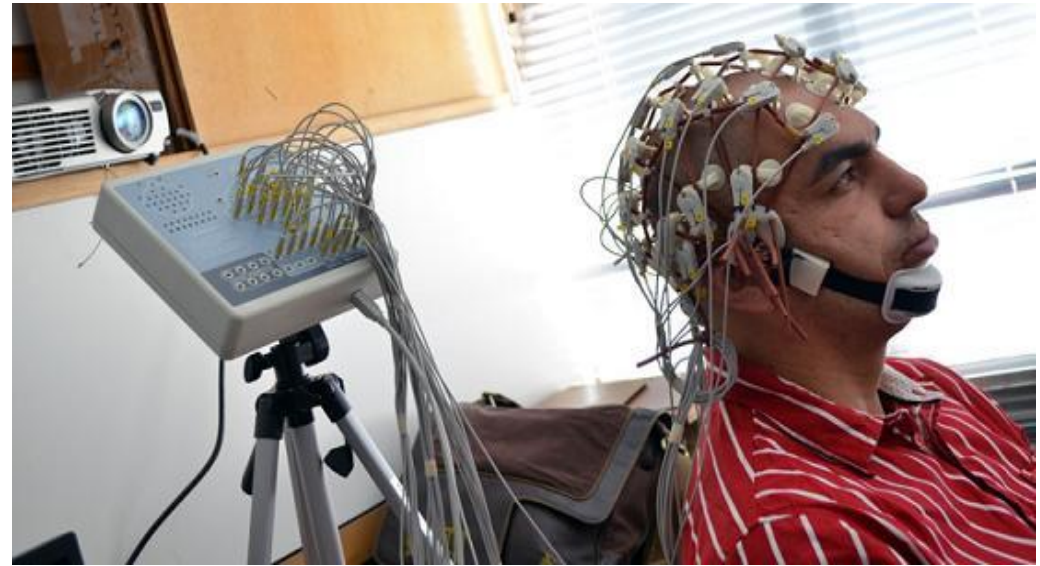
- types of mentions can be categorized into positive and negative themes, and can be calculated for net scores (positive mentions-negative mentions/total), and recommendation scores



Neuromarketing

- A new form of research that studies consumers' sensorimotor, cognitive, and affective response to marketing stimuli
 - Biometrics
 - Eye-tracking – observe eye movements
 - Heart rate, breathing rate
 - Skin response
 - Voice pitch
 - Neuroscience
 - Brain waves, brain activity – measure attention, memory, emotion
- Pro: accurate, used when consumers unaware of their own reactions or may not provide honest responses
- Con: very costly

Tools: Eyetracker, EEG, fMRI

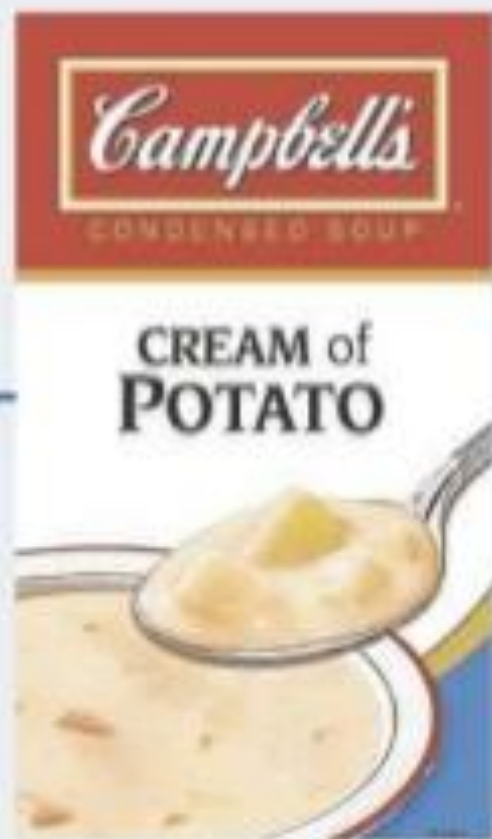


New and Improved . . . Labeling!

Campbell Soup used biometrics to analyze consumers' response to their label and changed the packaging to reflect their preferences revealed by the study.

New label

Old shelf label



CLASSIC
FAVORITES

*Cream of
Potato*

Campbell's
CONDENSED SOUP

The different varieties of soup were color-coded to help consumers distinguish them more easily.

Steam was added because people indicated they felt more emotionally engaged if the soup looked warm.

The spoon was removed. People thought it was unnecessary and had little emotional response to it.

The bowl was updated.

Eyetrack studies showed when the logo was placed at the top it drew too much attention and the red background also made all the labels look too similar.

Source: the company

Mystery shopping

- The use of individuals trained to experience and measure a customer service process, by acting as potential customers
 - Product recommendation:
 - Tell where and how retailers recommend products
 - Customer service experience:
 - Reveal how employees engage with customers, speed of service
 - Compliance:
 - Track compliance with company standards relating to service, product availability, price, store appearance
 - Retail audits:
 - Reports on store environments and point-of-sale communications

iPad 2:16 PM 44%

✕ SAMPLE Mystery Shopper Survey Submit

6 * How would you rate your first impression of the store?

Slide to select your answer:

Below Average Fantastic

7 * Did the Sales Associate ask if you needed help?

Yes No

8 * Did the Sales Associate inform you of any store promotions?

Yes No

9 * Were the price tags easy to understand?

Yes No

10 * Please track the amount of time you have to wait in the check-out line

Tap to start

11 * Was the check-out experience positive?

Yes No

12 Please enter any additional comments

Tap to answer

Observation – Pro & Con

- **Advantage**

- Measure **actual** rather than intended or preferred behavior
- No interviewer or reporting bias
- Reveal behavior patterns that respondents unaware or unable to communicate

- **Disadvantage**

- Time consuming
- Can be costly
- Reasons for the observed behavior may not be determined (underlying motives, beliefs, attitudes)

2. Survey

- Most widely used methods, best suited for gathering descriptive information
- Obtain consumers' knowledge, attitudes, preferences, or reported behaviors by asking consumers
- Pro:
 - Flexible, can obtain various kinds of info
 - Can be conducted by phone, mail, in person, online
- Con:
 - Answers may not accurate
 - Respondents may not remember or have never thought about what they do and why
 - Respondents give answers to please researcher, to appear smart
 - Not suited for sensitive, private issues

Question Design

- **Determine type of questions and answers**

- Closed-ended (choose the answer, easy to interpret).
Answers can be two choices (yes/no), multiple choice (never, sometimes, often, always), Likert scales (strongly disagree,...strongly agree), rating scales (1-5), ranking, checklist
- Open-ended (free to answer in their own words, harder to score but richer information, used in exploratory research, personal interview)

- **Length** - Short ones attract high response

- **Determine question order**

- Clear instructions how to answer
- Rapport questions to gain attention and stimulate interest
- Start with easy and non-threatening questions. Go from the general to the particular
- Put the most important items in first half
- Demographic and personal questions
 - Beginning: screen target respondents
 - Ending: when there are lots of difficult-to-answer questions

Closed question

Why don't you eat ice cream at Fictionals Ice Cream Parlour?
(Choose at least one answer.)

- ☐ I don't like the flavours
- ☐ It's too expensive
- ☐ The service is bad
- ☐ I don't like the ice cream
- ☐ It's too far from my house
- ☐ I don't know

Open-ended question

Why don't you eat ice cream at Fictionals Ice Cream Parlour?

I am lactose intolerant so I can't eat most ice creams, and it's really hard to find a store that offers good lactose-free ice cream. I've never heard of Fictionals but if I knew that they offered some, I would definitely try them out because I love ice cream!

Determine content/wording

Avoid ambiguity	Which brand of shampoo you use? What type of beverages do you prefer?
Avoid leading	"Don't you agree that soft drinks with sugar are bad for you?" Do you think the prices are cheaper at Asda than at Aldi?
Be specific	Avoid words like "regularly", "often", or "locally"
Avoid double-barreled	"On a scale from 1 to 10, how fast and reliable are Microsoft programs?"
Avoid double negatives	"Smoking in public places should not be abolished"
Avoid jargon, abbreviations	"Do you worry that you may have halitosis (bad breath)?"
Mutually exclusive options	"How many years have you worked in academia?" : 0-5, 6-10, 11-15, over 15." Not, "0-5, 5-10, 10-15..."
Difficult to answer	Amounts spent annually on specific product categories may not be known

Contact method

- Internet overcoming geographic boundary eliminates the postage, phone, interviewer, and data-handling costs, and thus internet-based survey is faster and usually cost much less. But the survey is self-administered.
- Personal survey allows researchers to obtain detailed information and ask follow-up questions, and test tangible stimuli but it's most expensive.

	Personal	Online
Flexibility	Excellent	Good
Quantity of data that can be collected	Excellent	Good
Control of interviewer effects	Poor	Fair
Control of sample	Good	Excellent
Speed of data collection	Good	Excellent
Response rate	Good	Good
Cost	Poor	Excellent

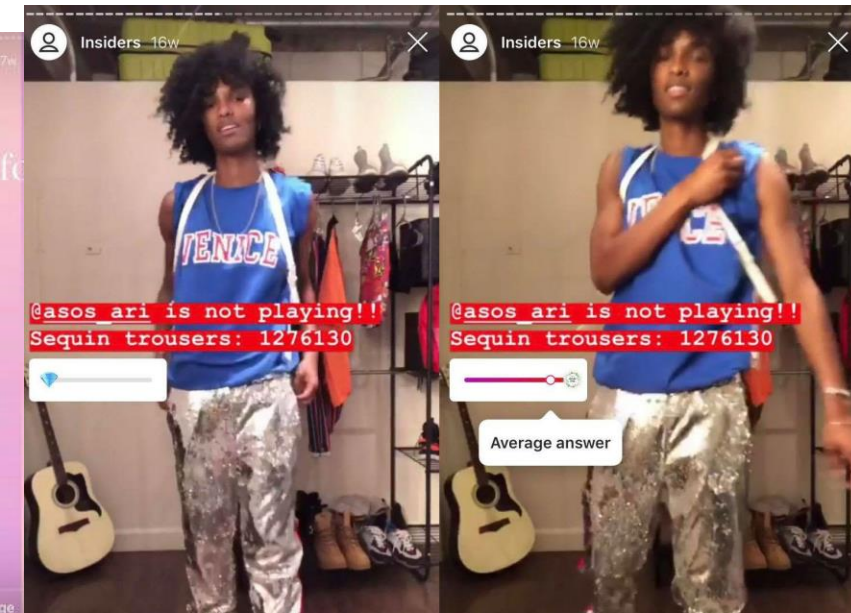
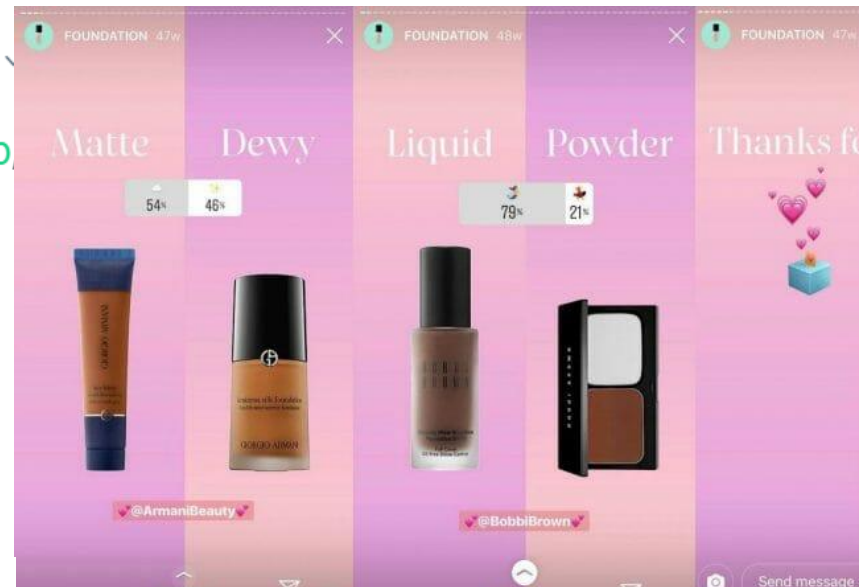
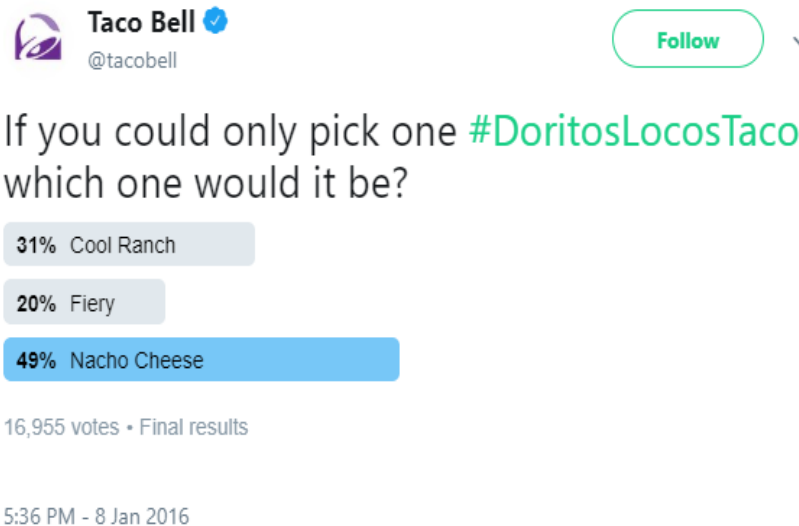
Personal Survey (Face-to-Face)

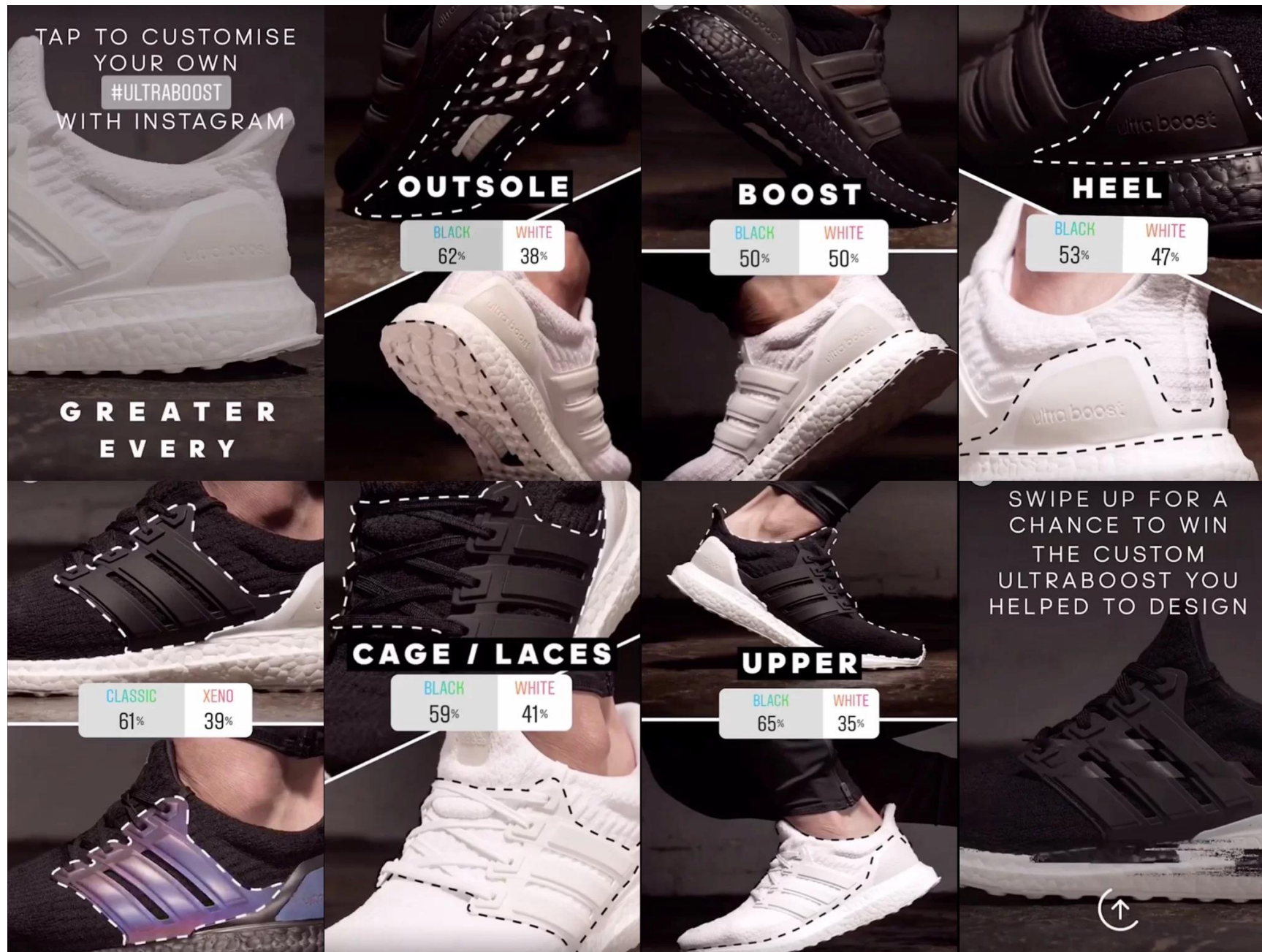
- Mall/street intercept, door-to-door
 - Not require the space renting
 - Suit for short questions, text-based, no media/ product test
 - New practice: use tablet instead of pen-paper task
- Central location test (CLT)
 - Appropriate for concept test, ad test, taste test
 - Ability to control the survey. Confidentiality.
 - Very costly
- Product placement test
 - Appropriate for product test
 - Very costly
 - Difficult to control - respondents may not follow the instruction



Social media survey campaign

- More interactive and engaging
- Polls (Facebook, Twitter, Instagram)
- Question Stickers- one-question polls for Stories (Instagram)
- Emoji slider (Instagram), allows followers to rate their products
- Call to action posts (Facebook): a poll with open-ended question
- Chatbot (Facebook, Line)





[Adidas](#) Running is giving consumers the opportunity to be part of the sneaker design process by voting for their favorite silhouette, which will then be released as a final product

The silhouette with the most votes, cast using IG's new functionality, will be released as an extremely limited-edition model. Only 50 pairs of the voted upon shoe will be created and everyone who votes will have the opportunity to win one of the numbered pairs.

3. Experiment

- A scientific investigation in which a researcher manipulates one cause variable, while controlling for other variables, and examines the effect on effect variables
- Experiments involve selecting matched groups of subjects, presenting them different levels of cause variables, and checking for differences in group responses.
 - Independent variables (cause): product characteristics, retail atmospherics (e.g. ambience, salesperson attractiveness)
 - Dependent variables (effect): choice, willingness to pay, purchase intention, satisfaction, attitudes
 - Group 1 taste the old flavor. Group 2 taste the new flavor. Each group rates the satisfaction with the product, without knowing what they were given
- Pro:
 - Determine the cause and effect of a study
- Con:
 - Costly, More difficult to do
 - Extra variable may affect the entire study

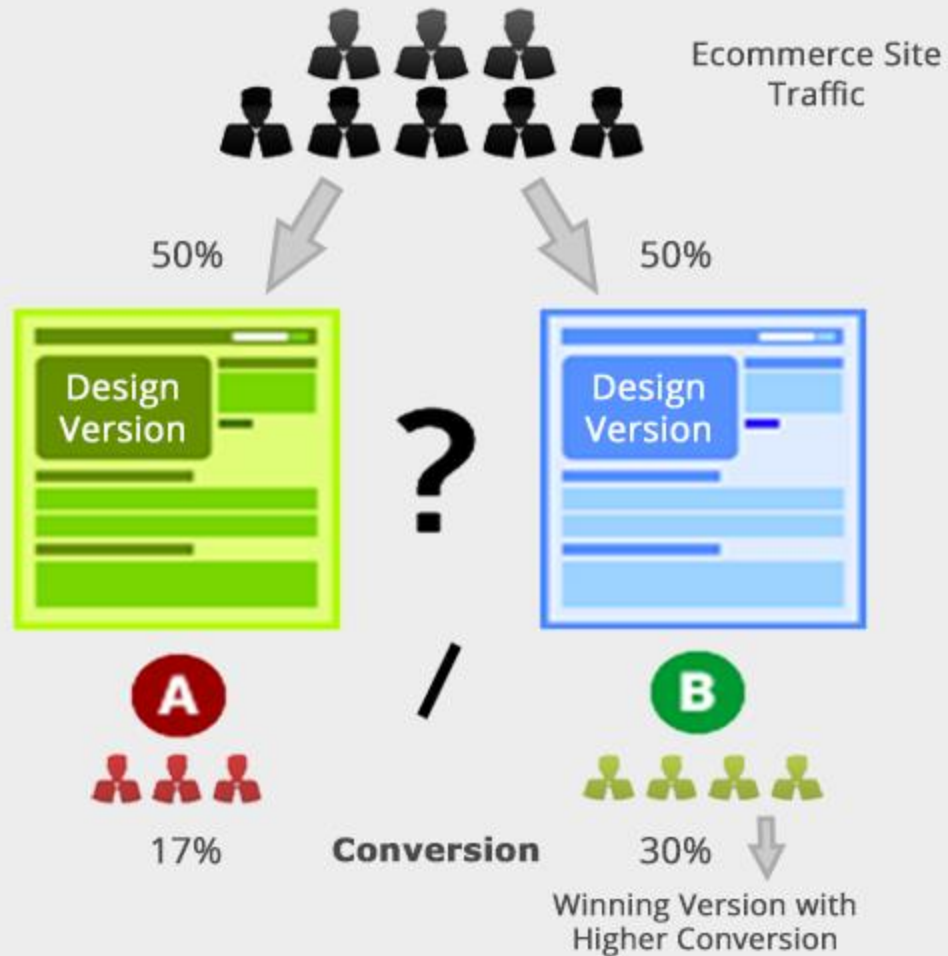
WITHOUT THE PROSTHESIS



WITH THE PROSTHESIS



A/B Testing for Better Ecommerce Conversion



Small Changes with a Huge Impact

Bing's experiments showed that slightly darker blues and greens in titles and a slightly lighter black in captions improved the users' experience. When rolled out to all users, the color changes boosted revenue by more than \$10 million annually.

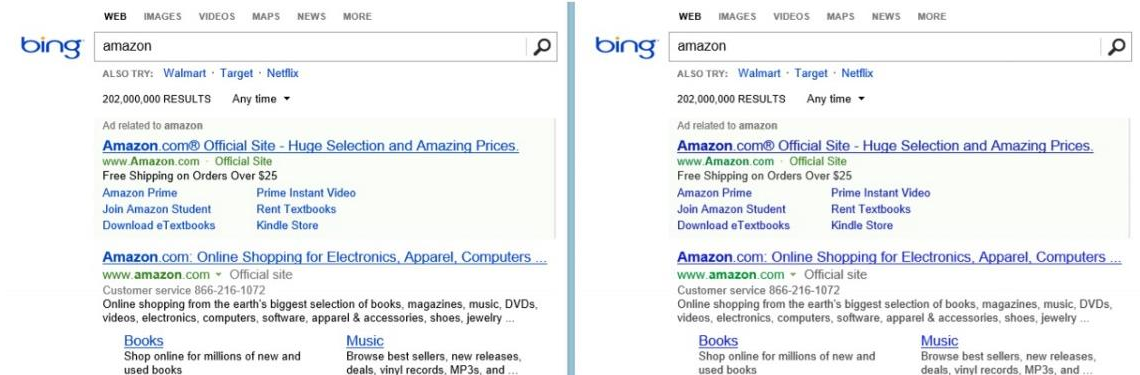
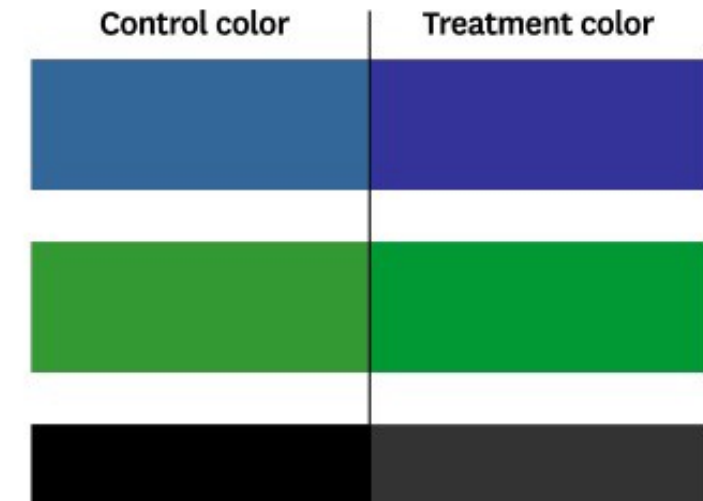


Figure 1: Font color experiment. Can you tell the difference?

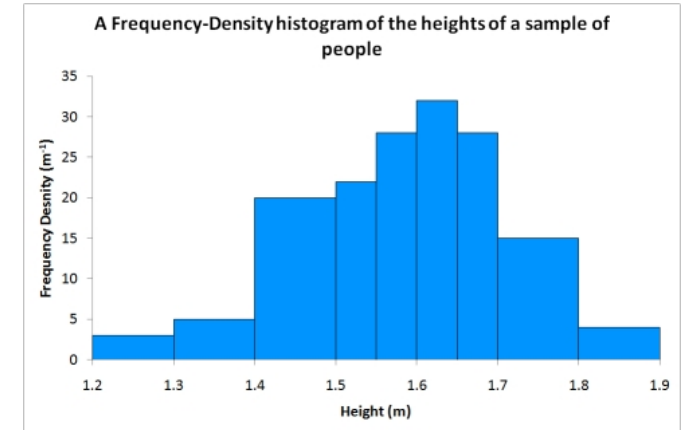
Step 4: Quantitative Data Analysis

The big picture of the data

- Frequency distribution
- Descriptive Statistics
 - Mean/Median/Mode
 - Range/Standard deviation

Tally for Discrete Variables: Pet

Pet	Count	Percent
Cat	47	47.00
Dog	53	53.00
N=	100	



Testing the relationship among variables

- Statistical testing (T-test): compare means
- Cross-tabulation (pivot table in excel)
- Correlation
 - auto insurance cost and accident (+)
- Regression
 - $y = ax_1 + bx_2 + e$

Tabulated statistics: Pet, Gender

Rows: Pet Columns: Gender

	Female	Male	All
Cat	35	12	47
	70	24	47
Dog	15	38	53
	30	76	53
All	50	50	100
	100	100	100

Cell Contents: Count
 % of Column

Step 4: Data Analysis

- Identifying and understanding the story you are trying to tell and the relationship you are looking to show.
- When analyzing data, search for patterns or interesting insights

TRENDS



Example:
Ice cream sales
over time

CORRELATIONS



Example:
Ice cream sales vs.
temperature

OUTLIERS



Example:
Ice cream sales in an
unusual region

DATA RELATIONSHIPS



NOMINAL COMPARISON

This is a simple comparison of the quantitative values of subcategories. Example: Number of visitors to various websites.



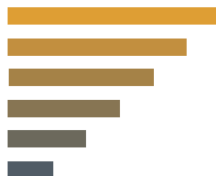
TIME-SERIES

This tracks changes in values of a consistent metric over time. Example: Monthly sales.



CORRELATION

This is data with two or more variables that may demonstrate a positive or negative correlation to each other. Example: Salaries according to education level.



RANKING

This shows how two or more values compare to each other in relative magnitude. Example: Historic weather patterns, ranked from the hottest months to the coldest.



DEVIATION

This examines how data points relate to each other, particularly how far any given data point differs from the mean. Example: Amusement park tickets sold on a rainy day vs. a regular day.



DISTRIBUTION

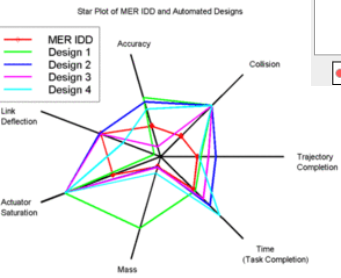
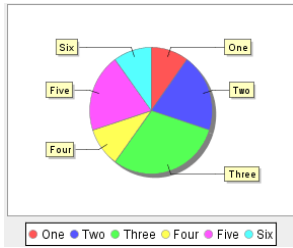
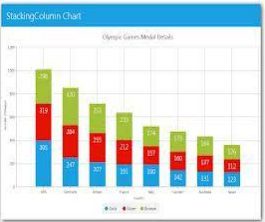
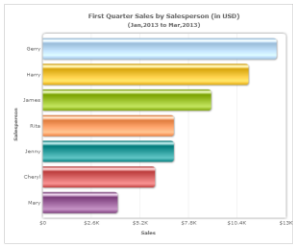
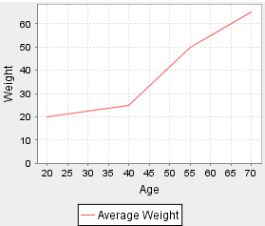
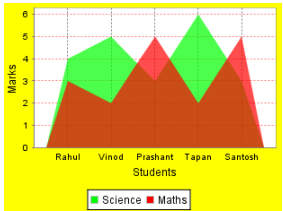
This shows data distribution, often around a central value. Example: Heights of players on a basketball team.



PART-TO-WHOLE RELATIONSHIPS

This shows a subset of data compared to the larger whole. Example: Percentage of customers purchasing specific products.

Step 5: Data Visualization



Type of chart	Purpose of the chart
Area chart	Emphasize the magnitude of change over time
Column chart, line chart, point chart	Show trends over time
Bar chart, column chart	Compare data, Show groups of related data
Stacked column chart	Show the parts that contribute to the total and compare change over time
Pie chart	Show the relationship of parts to the whole , Highlight proportions
Spider chart	Compare various aspects

Step 5: Insight and recommendation

- What does the data tell me about what I wanted to know?
- Why it happen
 - Ask questions and drill down
 - Create assumption and prove it (e.g. experiment)
 - Domain expertise
- How to action
 - Using what you know about customers, businesses, or markets to provide recommendations for how to move forward.

1. Survey
2. focus group
3. experiment/market test
4. observation
5. mystery shopper
6. database analysis
7. quantitative research
8. qualitative research
9. primary research
10. secondary data
11. internal records
12. marketing intelligence

Marketing Research Term <i>(Insert number from above table)</i>	Simple Definition/description
	Using 'pretend' customers to test the level of customer service
	Having a discussion with a group of about 10 consumers, for about 1.5 hours
	Reviewing the impact of our new advertising by examining customer purchases via their loyalty cards
	Phoning people at home to ask them a questionnaire
	Launching a new product in New York only to see how successful it will be
	Watching how our customers behave in the store
	A study undertaken to solve the marketing problem at hand
	Observing competitor's actions
	Statistical research – hard data and measurements
	Financial, sales and other data within the firm
	Existing data – from studies completed for some other purpose
	General term for more in-depth and richer research with a small number of consumers